

SkillVane IT Academy

Google Cloud Data Engineering + AI Fundamentals

Industry-Focused Training with Real-World Projects



ROLE

GCP Data Engineer



COURSE DURATION

3 Months



MODE

Online (English)



TRAINER

Shaik Saidhul

Website

YouTube

WhatsApp Community

LinkedIn

MODULE 1

GCP Cloud Basics

GCP Introduction

- ❖ The need for cloud computing in modern businesses.
- ❖ Key features and offerings of Google Cloud Platform (GCP).
- ❖ Overview of core GCP services and products.
- ❖ Benefits and advantages of using cloud infrastructure.
- ❖ Step-by-step guide to creating a free-tier account on GCP.

GCP Interfaces

Console

- ❖ Navigating the GCP Console
- ❖ Configuring the GCP Console for Efficiency
- ❖ Using the GCP Console for Service Management

Shell

- ❖ Introduction to GCP Shell
- ❖ Command-line Interface (CLI) Basics
- ❖ GCP Shell Commands for Service Deployment and Management

SDK

- ❖ Overview of GCP Software Development Kits (SDKs)
- ❖ Installing and Configuring SDKs
- ❖ Writing and Executing GCP SDK Commands

GCP Locations

Regions & Zones

- ❖ Understanding GCP Regions
- ❖ Selecting Regions for Service Deployment
- ❖ Impact of Region on Service Performance
- ❖ Exploring GCP Zones & Distributing Resources
- ❖ High Availability and Disaster Recovery Considerations

Importance

- ❖ Significance of Choosing the Right Location
- ❖ Global vs. Regional Resources
- ❖ Factors Influencing Location Decisions

GCP IAM & Admin

Identities & Roles

- ❖ Introduction to Identity and Access Management (IAM)
- ❖ Users, Groups, and Service Accounts
- ❖ Best Practices for Identity Management
- ❖ GCP IAM Roles Overview & Defining Custom Roles
- ❖ Role-Based Access Control (RBAC) Implementation

Policy & Resource Hierarchy

- ❖ Resource-based Policies
- ❖ Understanding and Implementing Organization Policies
- ❖ Auditing and Monitoring Policies
- ❖ GCP Resource Hierarchy Structure
- ❖ Managing Resources in a Hierarchy
- ❖ Organizational Structure Best Practices

MODULE 2



Linux Basics on Cloud Shell

- ❖ Getting started with Linux & Installation
- ❖ Basic Linux Commands & Cloud shell tips
- ❖ File and Directory Operations (ls, cd, pwd, mkdir, rmdir, cp, mv, touch, rm, nano)
- ❖ File Content Manipulation (cat, less, head, tail, grep)
- ❖ Text Processing (awk, sed, cut, sort, uniq)
- ❖ User and Permission related (whoami, id, su, sudo, chmod, chown)

MODULE 3



Python for Data Engineer

- ❖ **Data Types:** Strings, Operators, Numbers (Int, Float), Booleans
- ❖ **Data Structures:** Lists, Tuples, Dictionaries, Sets
- ❖ **Python Programming Constructs:**
 - ❖ if, elif, else statements, for loops, while loops
 - ❖ Exception Handling & File I/O operations
- ❖ **Modular Programming:** Functions & Lambda Functions, Classes

MODULE 4

 GCP Data Engineering Tools
Google Cloud Storage

- ❖ Overview of Cloud Storage as a scalable and durable object storage service.
- ❖ Understanding buckets and objects in Cloud Storage.
- ❖ Use cases for Cloud Storage, such as data backup, multimedia storage, and website content
- ❖ Labs: using console & CLI to do below
 - Creating and managing Cloud Storage buckets.
 - Uploading and downloading objects to and from Cloud Storage.
 - Setting access controls and permissions for buckets and objects.
- ❖ Data Transfer and Lifecycle Management
- ❖ Versioning and Object Versioning
- ❖ Integration with Other GCP Services
- ❖ Monitoring and logging for Cloud Storage operations.

Cloud SQL

- ❖ Introduction to Cloud SQL
- ❖ Creating and Managing Cloud SQL Instances
- ❖ Configuring database settings, users, and access controls.
- ❖ Connecting to Cloud SQL instances using Cloud SQL studio, Shell, Workbenches
- ❖ Importing and exporting data in Cloud SQL.
- ❖ Backups and High Availability
- ❖ Integration with Other GCP Services
- ❖ Managing database user roles and permissions.
- ❖ Introduction to DMS
- ❖ End to End Database migration Project
 - Manual: Export and Import method
 - Automation: Cloud SQL DMS method

BigQuery (SQL development)

- ❖ Introduction to BigQuery & Architecture
- ❖ Use cases for BigQuery in business intelligence and analytics.
- ❖ Various method of creating table in BigQuery (Native table and External Tables)
- ❖ BigQuery Data Sources and File Formats
- ❖ Working with complex data types (json data, nested, repeated and array data)
- ❖ Data Integration and Export
 - Loading data into BigQuery from Cloud Storage, Cloud SQL, and other sources.
 - Exporting data from BigQuery to various formats.
 - Real-time data streaming into BigQuery.
- ❖ Configuring access controls and permissions in BigQuery.
- ❖ BigQuery Views: Views, Materialized Views, Authorized Views
- ❖ Optimization techniques in BigQuery
- ❖ BigQuery Slots - on demand, flat-rate, flex-slots
- ❖ Case Study-1 : implement a real-world analytics data platform for Spotify
- ❖ Case Study-2: Enterprise Social Media analytics platform

DataProc (Pyspark Development)

- ❖ Introduction to Hadoop and Apache Spark
- ❖ Understanding the difference between Spark and MapReduce
- ❖ What is Spark and Pyspark.
- ❖ Understanding Spark framework and its functionalities
- ❖ Overview of DataProc as a fully managed Apache Spark and Hadoop service.
- ❖ Cluster Creation and Configuration
 - Creating and managing DataProc clusters.
 - Configuring cluster properties for performance and scalability.
 - Preemptible instances and cost optimization.
- ❖ Learning Pyspark:
 - How to read from multiple data sources - csv, text, json, parquet, database table, BigQuery tables
 - How to perform multiple transformations
 - How to write to multiple targets - csv, text, json, parquet, database table, BigQuery tables
- ❖ Running Jobs on DataProc
 - Submitting and monitoring Spark and Hadoop jobs on DataProc.
 - Use of initialization actions and custom scripts.
 - Job debugging and troubleshooting.
- ❖ Case study-1: Data Cleaning of Employee Travel Records
- ❖ Case study-2: Processing real-time patient health data
- ❖ Case study-3: Creating a pyspark job to support ML model creations

DataFlow (Apache Beam development)

- ❖ Introduction to DataFlow
- ❖ Use cases for DataFlow in real-time analytics and ETL.
- ❖ Understanding the difference between Apache Spark and Apache Beam
- ❖ How Dataflow is different from Dataproc
- ❖ Learning Apache Beam:
 - How to read from multiple data sources - csv, text, json, parquet, database table, BigQuery tables
 - How to perform multiple transformations
 - How to write to multiple targets - csv, text, json, parquet, database table, BigQuery tables
- ❖ Case study-1: Template method of creating pipelines
- ❖ Case study-2: E-commerce Transaction Processing
- ❖ Case study-3: End to End Streaming Pipeline using Apache beam with Dataflow, Python app, PubSub, BigQuery, GCS

Cloud Pub/Sub (Streaming)

- ❖ Introduction to Pub/Sub
- ❖ Understanding the role of Pub/Sub in event-driven architectures.
- ❖ Key Pub/Sub concepts: topics, subscriptions, messages, and acknowledgments.
- ❖ Creating and Managing Topics and Subscriptions
 - Using the GCP Console to create Pub/Sub topics and subscriptions.
 - Configuring message retention policies and acknowledgment settings.
- ❖ Publishing and Consuming Messages
 - Writing and deploying code to publish messages to a topic.
 - Implementing subscribers to consume and process messages from subscriptions.
- ❖ Case study-1: Streaming use-case using Dataflow

Cloud Composer (DAG Development)

- ❖ Introduction to Composer/Airflow & Overview of Airflow Architecture
- ❖ Use cases for Composer in managing and scheduling workflows.
- ❖ Creating and Managing Workflows
 - Creating and configuring Composer environments.
 - Defining and scheduling workflows using Apache Airflow.
 - Monitoring and managing workflow executions.
- ❖ Integration with Data Engineering Services
 - Orchestrating workflows involving BigQuery, DataFlow, and other services.
 - Coordinating ETL processes with Composer.
 - Integrating with external systems and APIs.
- ❖ Error Handling and Troubleshooting
 - Handling errors and retries in Composer workflows.
 - Debugging and troubleshooting failed workflow executions.
 - Logging and monitoring for Composer workflows.
- ❖ Level-1-DAG: Orchestrating the BigQuery pipelines
- ❖ Level-2-DAG: Orchestrating the DataProc pipelines
- ❖ Level-3-DAG: Orchestrating the Dataflow pipelines
- ❖ Deploy DAGs: Implementing CI/CD in Composer Using Cloud Build and GitHub

MODULE 5



Databricks on GCP

- ❖ **a. What is Lakehouse Architecture?**
- ❖ **b. Difference Between:** Data Lake, Data Warehouse, Data Lakehouse
- ❖ **c. Introduction to Databricks**
 - Overview of Databricks
 - Databricks Architecture
 - Setting up a Databricks Workspace
- ❖ **d. Unity Catalog - Unified Data Governance**
 - Introduction to Unity Catalog
 - Core Concepts: Metastore, Catalog, Schema, Tables, Volumes, Functions, Models
 - External Data Access: a. Storage Credentials, b. External Locations (S3, GCS, ADLS)
 - Lakehouse Federation - Foreign Catalog, Connection
 - Delta Sharing - Share, Recipient, Provider
- ❖ **e. Databricks Clusters**
 - Introduction to Clusters, Types of Clusters, Cluster Modes
- ❖ **f. DBUtils Commands** - File handling, Notebook Widgets, Secrets
- ❖ **g. Introduction to Delta Lake**
 - What is Delta Lake?
 - Creating Delta Lake Tables - Managed Tables, External Tables
 - Understanding Delta Lake Table Creation - Delta Log, Parquet Data Files
 - Advanced Delta Lake Features: a. Versioning, b. Time Travel, c. Compacting, d. Liquid clustering, e. Vacuum
- ❖ **h. Databricks Jobs & Pipelines**
 - Creating a Job, Scheduling Jobs, Setting Parameters
 - Managing Dependencies, Setting up Alerts
- ❖ **i. Case study-1:** Structured Streaming with Databricks
- ❖ **j. Case study-2:** Incremental Data Loading with Auto Loader

MODULE 6

 Complementary Services**Data Fusion (complementary)**

- ❖ Introduction to Data Fusion & Overview of Data Fusion as a fully managed data integration service.
- ❖ Use cases for Data Fusion in ETL and data migration.
- ❖ Building Data Integration Pipelines
 - Creating ETL pipelines using the visual interface.
 - Configuring data sources, transformations, and sinks.
 - Using pre-built templates for common integration scenarios.
- ❖ Integration with GCP and External Services (BigQuery, Cloud Storage)
- ❖ Case study-1: End to End pipeline using Data fusion with Wrangler, GCS, BigQuery

Cloud Functions (complementary)

- ❖ Cloud Functions Introduction & Setting up Cloud Functions in GCP
- ❖ Event-driven architecture and use cases
- ❖ Writing and deploying Cloud Functions
- ❖ Triggering Cloud Functions: HTTP triggers, Pub/Sub triggers, Cloud Storage triggers
- ❖ Monitoring and logging Cloud Functions
- ❖ Usecase-1: Loading the files from GCS to BigQuery as soon as it is uploaded.

Terraform (complementary)

- ❖ Terraform Introduction & Installing and configuring Terraform.
- ❖ Infrastructure Provisioning
 - Terraform basic commands: Init, plan, apply, destroy
- ❖ Labs: Create Resources in Google Cloud Platform (GCS buckets, Dataproc cluster, BigQuery Datasets and tables)
- ❖ And more resources as needed

MODULE 7

 AI Fundamentals for GCP Data Engineers

- ❖ **Introduction to Artificial Intelligence & Generative AI**
 - Understanding AI, ML, Deep Learning, Generative AI, and LLMs.
 - Real-world AI use cases in Data Engineering and Analytics.
- ❖ **Machine Learning Fundamentals**
 - Supervised, Unsupervised, and Reinforcement Learning.
 - Training, validation, testing, and model evaluation concepts.
- ❖ **Prompt Engineering for Data Engineers**
 - Designing effective prompts for Gemini, ChatGPT, and Vertex AI.
 - Using AI for SQL generation, code generation, documentation, and data analysis.
- ❖ **Google Vertex AI Fundamentals**
 - Overview of Vertex AI ecosystem.
 - Working with foundation models, Model Garden, and Gemini APIs.
- ❖ **AI-Powered Data Pipelines**
 - Integrating AI services with BigQuery, GCS, Dataflow, and Dataproc.
 - Building intelligent ETL/ELT pipelines using AI-driven transformations.

Complementary End-to-End Projects:

1. Healthcare project on GCP of 8+ hours
2. Road Traffic project on Databricks of 5+ hours

By the End of the course What Students can Expect

Proficient in SQL Development

- ✓ Mastering SQL for querying and manipulating data within Google BigQuery and Cloud SQL.
- ✓ Writing complex queries and optimizing performance for large-scale datasets.
- ✓ Understanding schema design and best practices for efficient data storage.

Apache Beam Development Mastery

- ✓ Creating data processing pipelines using Apache Beam.
- ✓ Understanding the concepts of parallel processing and data parallelism.
- ✓ Implementing transformations and integrating with other GCP services.

Notebooks, Workflows with Databricks

- ✓ Understand how to build and manage data pipelines using Databricks and Delta Lake.
- ✓ Efficiently query and analyze large datasets with Databricks SQL and Apache Spark.
- ✓ Implement scalable workflows and optimize performance within Databricks.

ML & AI ready

- ✓ Students will understand how modern AI and Generative AI integrate with Google Cloud Data Engineering solutions and will be able to build AI-enabled data platforms using BigQuery, Vertex AI, and Gemini.

Pyspark Development Skills

- ✓ Proficiency in using PySpark for large-scale data processing on Google Cloud.
- ✓ Developing and optimizing Spark jobs for distributed data processing.
- ✓ Understanding Spark's RDDs, Dataframes, and transformations for data manipulation.

DAG Creations with Cloud Composer

- ✓ Designing and implementing Directed Acyclic Graphs (DAGs) for orchestrating workflows.
- ✓ Using Cloud Composer for workflow automation and managing dependencies.
- ✓ Developing DAGs that integrate various GCP services for end-to-end data processing.

Architecture Planning & Certification

- ✓ Proficient in architecting end-to-end data solutions on GCP.
- ✓ Understanding the principles of designing scalable, reliable, and cost-effective data architectures.
- ✓ Prepare for the Google Cloud Professional Data Engineer (PDE) and Associate Cloud Engineer (ACE) certifications through a combination of theoretical knowledge and hands-on experience.

Course Highlights



Hands-on Labs & Real-world Use Cases



Live Q&A Sessions



Course Completion Certificate



Access to Study Materials & Code Repository



Industry-Standard Best Practices

The course will empower students with practical skills in SQL, PySpark, Apache Beam, DAG creations, and architecture planning, ensuring they are well-prepared to tackle real-world data engineering challenges and successfully obtain GCP certifications.

Thank You.